

Arithmetic Sequences

For an arithmetic sequence,

$$a_n = a_1 + (n-1)d, \quad a_n = \frac{a_{n-1} + a_{n+1}}{2}, \quad S_n = \frac{n}{2}(a_1 + a_n) = \frac{n}{2}(2a_1 + (n-1)d).$$

1. Given

$$a_{18} = -26, \quad a_8 - a_{25} = -29.$$

Find a_{25} and d .

2. Given

$$a_{27} = -22, \quad a_5 + a_{22} = -2.$$

Find a_{16} and d .

3. Given

$$a_7 = -14, \quad a_{29} - a_{14} = 20.$$

Find a_{18} and d .

4. Given

$$a_{15} = 1, \quad a_{27} + a_2 = -7.$$

Find a_8 and d .

5. Given

$$a_{15} = -24, \quad a_3 - a_{16} = -27.$$

Find a_{24} and d .

6. Given

$$a_{27} = 21, \quad a_9 - a_{23} = -25.$$

Find a_{49} and d .

Additional Problems

1. What is the sum of the two-digit numbers ending in 5?

2. Given

$$a_{10} = 3, \quad a_{18} = 11.$$

Find a_n , d , and S_{20} .

3. Given

$$S_5 = 15.$$

We know that a_3 is five times a_1 . Find a_n and d .

Geometric Sequences

For a geometric sequence,

$$a_n = a_1 q^{n-1}, \quad a_n^2 = a_{n-1} a_{n+1}, \quad S_n = a_1 \frac{q^n - 1}{q - 1}, \quad q \neq 1.$$

1. Given

$$a_7 = \frac{1}{8}, \quad q = \frac{1}{2}.$$

Find $a_1 + a_2$.

2. Given

$$a_4 = 108, \quad a_{10} = 78432.$$

Find q and a_1 .

3. Given

$$a_7 = 256, \quad q = -2.$$

Find a_2 .

4. Given

$$a_4 = \frac{1}{9}, \quad a_9 = \frac{1}{729}, \quad q > 0.$$

Find a_1 .

5. Given

$$a_2 = 8, \quad a_{10} = 2048, \quad q > 0.$$

Find a_1 .

6. Given

$$a_3 = \frac{1}{3}, \quad a_{11} = \frac{1}{19683}, \quad q > 0.$$

Find a_1 .

7. Given

$$a_4 = 32, \quad a_{10} = 2048, \quad q > 0.$$

Find a_1 .

8. Given

$$a_4 = 54, \quad a_{12} = 354294, \quad q > 0.$$

Find a_1 .

9. Given

$$a_2 = -\frac{7}{2}, \quad q = -\frac{1}{2}.$$

Find a_8 .

Additional Problems

1. a_6 is 243 times a_1 , and

$$a_2 + a_4 = 60.$$

Find a_1 and q .

2. a_4 is four times a_2 , and

$$a_3 a_5 = 16.$$

Find a_1 and q .

3. Given

$$S_3 = 21, \quad S_6 = 189.$$

Find a_1 , q , and S_8 .

Financial Problems

1. We bought a product for 18 million HUF. Its value decreases by 1.1% each month. How much will it be worth 20 months later?
2. We deposit 1 million HUF at an annual interest rate of 5%. How much money will we have after 10 years with annual compounding? How much will we have after 10 years with monthly compounding?
3. At an interest rate of 6%, how many years are needed at least for our money to double? Assume annual compounding.